Protecting the `main` branch in GitHub

As noted in the class about Git Flow, all development occurs on *feature branches*; no developer ever works on (or pushes to) the `main` branch.

To enforce this practice (and avoid mistakes!), follow the instructions below to tell GitHub to protect the main branch:

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1. From your repository's home page on GitHub, click *Settings*;
2. In the left menu, click *Branches*;
3. To the right of *Branch protection rules*, click *Add rule* button;
4. Under *Branch name pattern*, type `main`;
5. Under *Protect matching branches*,
6. click *Require pull request reviews before merging*
7. click *Include administrators*
8. At the bottom of the page, click green *Create* button.

> **Note**
> The above comes from the GitHub documentation for protecting the main branch.

Then, if anyone ever tries to push to `main`, perhaps by accident, they'll get this error:

```
$ git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 738 bytes | 738.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
remote: error: GH006: Protected branch update failed for refs/heads/main.
remote: error: At least 1 approving review is required by reviewers with write access.
To github.com:https://github.com/aninja/activity-git-flow.git
 ! [remote rejected] main -> main (protected branch hook declined)
error: failed to push some refs to 'github.com:https://github.com/aninja/activity-git-flow.git'
```

If a developer has accidentally worked on `main`, committing to `main` as they go, this error can be quite discouraging!
The solution is to create a branch, now.
(Yes, they should have done it *before* doing this work, but now it is the only way out of this mess.)

```
$ git branch feature
$ git switch feature
Switched to branch 'feature'
$ git push origin feature
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 738 bytes | 738.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature' on GitHub by visiting:
remote:      https://github.com/aninja/activity-git-flow/pull/new/feature
remote:
To https://github.com/aninja/activity-git-flow.git
 * [new branch]      feature -> feature
```

Then they can create a pull request using the link provided.

Before they start work on a new feature, they should create a new branch for the new feature:

```
$ git switch main
$ git pull origin main
$ git branch feature2
$ git switch feature2
```